



PROGRAMMING FUNDAMENTALS

CS – 101

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VARIABLES, EXPRESSIONS & STATEMENTS

- **VALUE:**

- Basic entity, a program works with
- Defined by a letter or number, for instance:
 - 1, 24, 309
 - “Hello World!”, “PFUN is fun!”
 - 2.99, 7.5, 5.0
- Belongs to a specific category called type

VARIABLES, EXPRESSIONS & STATEMENTS

- TYPE:

- Represents the category to which a value belongs to
- Similar to natural languages
 - 1, 24, 309 --- > are all whole numbers, i.e. Integers (Python keyword: **int**)
 - "Hello World!", "PFUN is fun!" --- > are string of characters, i.e. String (Python keyword: **str**)
 - 2.99, 7.5, 5.0 --- > are floating-point numbers, i.e. Float (Python keyword: **float**)
- To find the type of any specified value, use the keyword **type**
i.e. `type(some_value)`
- To print the type of the value on screen use **print** statement with the **type** statement
i.e. `print(type(some_value))`

COMMON MISTAKES

- When specifying a string value, use quotation marks, without which it'll consider it as a variable
i.e. `print("some_text")` -- > output: some_text
- Numeric values written in quotation marks are considered as string values and not numeric
i.e. `print("33 – 5")` -- > output: 33 – 5
- Numbers cannot be separated by commas like done in mathematics
i.e. `print(1,234,567)` -- > output: 1 234 567 (three values)

VARIABLE

- A label or name that can refer to a value
- To create new variables and give them a value, assignment statement is used
- For example
 - msg = "I am a student of CS-101"
 - Assigns a string statement "I am a student of CS-101" to variable "msg"
 - n = 17
 - Assigns an integer value, 17 to variable "n"
 - pi = 3.1415926
 - Assigns a floating-point number, 3.1415926 to variable "pi"
- Cannot assign a numeric value with a leading zero to a variable
i.e. zipcode = 02492 --- > will give an error

VARIABLE NAMES

- Selected by the programmer
- Can contain both letters and numbers
- Must begin with a letter
- Underscore can be used in between names (my_name, my_class_fun)
- No blank space between name (*my name* is illegal)
- Case sensitive
- Can begin with either uppercase or lowercase letters but uppercase is not preferable

VARIABLE NAMES (COMMON MISTAKES)

- What is wrong with the following assignment statements?
 1. 76trombones = "Big Parade!"
 2. more@ = 2000
 3. class = "PFUN"
 4. movie name = "Jurassic Park"

KEYWORDS

- Pre-defined names to perform different functionalities by the programming language
- Cannot use keywords as variable names
- List of keywords:

and	del	from	not	while
as	elif	global	or	with
assert	else	if	pass	yield
break	except	import	print	
class	exec/nonlocal	in	raise	
continue	finally	is	return	
def	for	lambda	try	

OPERATORS AND OPERANDS

- Operators are special symbols that represent computations, like addition, multiplication,
- Operands are the entities or values on which the operator is applied to perform the computation

SYNTAX	OPERATORS	OPERANDS	COMPUTATION PERFORMED
20 + 32	+	20, 32	Addition
Hour – 1	–	Hour, 1	Subtraction
Speed * 5	*	Speed, 5	Multiplication
Distance/Time	/	Distance, Time	Division
Base**3	**	Base, 3	Exponent

- Floor division: When both inputs are Integers, the result is expected to be integer, so it rounds down the value to the nearest Integer. If one of the operands is a Float, then the result will be rounded down to nearest integer, but it will be represented as Float. The operator is represented by //

EXPRESSIONS AND STATEMENTS

- Expression is a combination of values, variables, and operators
 - 17
 - x
 - $x + 18$
- Statement is a piece of code that the interpreter can execute as in print or assignment statements, as discussed earlier
 - `print(17)`
 - `x = 30`
 - `print(x + 18)`

ORDER OF OPERATIONS

- All operators must be defined by their precedence order
- Python uses **PEMDAS** rule
- Parenthesis
 - Have the highest priority
 - Can be used to prioritize operations, as well as for clearer representation of code
 - Examples:
 - $2 + (3 - 1) = ?$ Output = 4
 - $(1 + 1) ** (5 - 2) = 2 ** 3 = ?$ Output = 8

ORDER OF OPERATIONS

- Exponent

- The next priority after parenthesis

- Examples:

➤ $2 ** 1 + 1 = 2 + 1 = ?$ Output = 3

➤ $3 * 1 ** 3 = 3 * 1 = ?$ Output = 3

- Multiplication and Division

- Same precedence

- If both present, it solves left to right

- Examples:

➤ $2 * 3 - 1 = 6 - 1 = ?$ Output = 5

➤ $6 + 4 / 2 = ?$ Output = 8

➤ $24 / 2 * 3 = ?$ Output = 36

➤ $3 * 3 / 2 = ?$ Output = 4.5

➤ $24 / 3 / 3 = ?$ Output = 2.66

ORDER OF OPERATIONS

- Addition and Subtraction

- Both has the same precedence
- The least priority of all the mathematical calculations
- If both present, it solves them from left to right
- Examples:

➤ $55 + 4 - 32 = ?$ Output = 27

➤ $23 - 31 + 100 = ?$ Output = 92

STRING OPERATIONS

- Cannot perform mathematical operations on Strings
- It is illegal to have statements like:
 - `'2' - '1'`
 - `"eggs" / "easy"`
- However, `+` and `*` operators work on Strings but in a different way
- The `+` operator is not used to add the two strings, instead, it concatenates them, i.e. join the two strings
 - `First = "Favorite"`
 - `Second = "Food"`
 - `print(First + Second)` --- > Output: FavoriteFood
- The `*` operator is used for repetition
 - `print("Spam" * 3)` --- > Output: SpamSpamSpam

COMMENTS

- Used to explain the process so that others can understand
- Written in the form of notes using Natural Language to explain
- These notes are called comments
- Comment line is not executed by the machine
- A single line comment is preceded by # symbol
- Single line comment can be written on a separate line or after the instruction
- Multiple line comments is specified by three single quotations before and after the desired piece of code

'''

Some code

'''

EXERCISE

- Assume that we execute the following assignment statements:

width = 17

height = 12.0

delimiter = '.'

- For each of the following expressions, write the value of the expression and the type of the resultant values

EXERCISE (continued)

Expression	Value	Type
width//2.0		
width/2		
width/2.0		
height/3		
height//3		
1 + 2 * 5		
delimiter * 4		

EXERCISE (continued)

Expression	Value	Type
width//2.0	8.0	float
width/2	8.5	float
width/2.0	8.5	float
height/3	4.0	float
height//3	4.0	float
1 + 2 * 5	11	int
delimiter * 4		str

SELF-PRACTICE EXERCISES

- Hakerrank Link: <https://hr.gs/cs102Fall2022SPTaskCh2>
- These questions will be available until next Friday, September 9th, 2022, so do work on them, as well as all the other practice problems being shared by the other faculty and staff.
- These are Self Practice Tasks so they will not be graded. If you have any “questions about the questions” 😊 you can come see me during my office hours:
 - Tuesdays/Thursdays: 2:00pm – 3:00pm
 - Office: W-230